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Embedded Interdependence: Real Output Linkages Between Global and Dependent Economies

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Abstract

Over the past three decades, the economic ties linking the United States, China, Germany, and the Visegrád Four (V4) have grown exceptionally tight. While this deep integration has driven historical growth, it has also left these economies highly exposed to sudden global shocks. To understand exactly how these shocks travel, this study applies the Diebold-Yılmaz rolling-window connectedness framework to trace real output spillovers between 1995 and 2023. Beyond simply measuring general interdependence, the research maps out a distinct structural chain reaction. It demonstrates how large-scale shocks originate in systemic drivers like the US and China, pass directly through Germany as a central industrial hub, and ultimately land on the hyper-open economies of the V4. The data reveals a massive degree of macroeconomic overlap, backed by a Total Connectedness Index (TCI) of 83.78%. This confirms that output shocks cross borders rapidly, heavily dictating the local business cycles of the Central European region. Consequently, because global fluctuations essentially drive V4 economic performance, policymakers must actively build domestic buffers to withstand future international volatility.

1 Introduction

Traditional neoclassical growth theories [1] [2] treated national economies as largely self-contained units. However, the rapid acceleration of trade and financial globalization has effectively dismantled this closed-economy assumption. In the modern economic landscape, cross-border interdependence is the norm, and national business cycles are heavily shaped by external shocks [3] [4]. Because of this, mapping the exact sources and directions of growth spillovers is now a core focus of macroeconomic research. Within this highly integrated global system, the relationship between the United States, China, Germany, and the Visegrád Four (V4) stands out as a critical transmission channel.

These seven economies form a fascinating cross-section of the global market. While they vary drastically in their historical backgrounds, political structures, and absolute economic size, their integration since the 1990s has tied their economic fates together. The post-Cold War transformation of the V4, China’s industrial explosion, Germany’s anchor role in Europe, and the persistent global weight of the US have forged deep structural links. These connections go beyond simple bilateral trade, extending deeply into shared global value chains and synchronized business cycles [5] [6]. As these ties have solidified, tracking exactly how an output shock in one region ripples through the rest has become crucial for designing sound macroeconomic policy.

The underlying vulnerability within this network becomes obvious when looking at trade openness. According to World Bank data, trade in the V4 countries regularly exceeds 150% of their GDP, a clear reflection of their heavy reliance on European supply chains [7]. Germany operates as a highly open export engine with trade accounting for roughly 80-90% of its GDP. The US and China, on the other hand, show much lower relative openness, usually hovering between 20% and 35%. Figure 1 illustrates these historical differences across the analyzed economies. Because of their massive absolute size, the US and China dictate global demand and generate systemic shocks. This creates a clear, asymmetric hierarchy: the US and China act as global originators, Germany serves as the primary transmission hub, and the hyper-open V4 economies function as the ultimate receivers of these shocks.

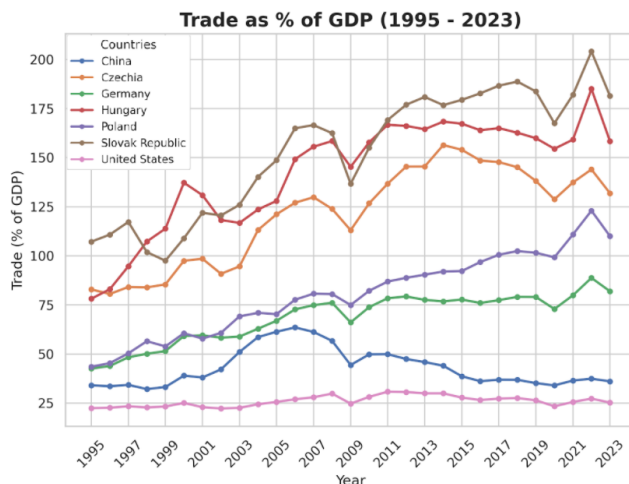


Figure 1: Trade Openness Divergence: V4 vs. Major Economies (1995–2023)

While these relationships are widely acknowledged, empirical work rarely measures their real output connectedness within a single joint framework. Much of the existing literature looks either at financial market spillovers or isolated bilateral trade relations, often missing the broader systemic interaction in real economic activity [8] [9]. The network-based methodology developed by Diebold and Yilmaz [5] [10] [11] [12] is highly popular in the financial literature, but applying it to real GDP dynamics across such structurally different regions remains largely unexplored.

To fill this analytical void, this paper investigates the output connectedness among the US, China, Germany, and the V4 from 1995 to 2023. By applying the generalized Diebold-Yılmaz framework, the research seeks to:

- Measure the overall magnitude and changing nature of output spillovers across these regions;
- Pinpoint which economies act as the primary drivers of output shocks and which are the most vulnerable;
- Trace the exact directional linkages within and between these blocs;
- Extract practical policy lessons for macro-financial stability and regional coordination.

To properly understand the V4’s position in this network, it is helpful to look at the Varieties of Capitalism (VoC) framework [13]. Nölke and Vliegenthart [14] classify the Visegrád nations as “dependent market economies” (DMEs). Rather than relying on domestic institutional coordination, DMEs are driven by foreign direct investment, export-heavy manufacturing, and the deep integration of multinational firms, especially in the automotive and electronics sectors. Their financial systems reflect this as well, typically featuring a high share of foreign bank ownership [15], which sets them apart from the more domestically anchored model of Slovenia or the highly liberalized Baltic states [16] [17]. Bohle and Greskovits [18] describe this V4 model as “embedded neoliberalism”, combining market openness with EU-driven institutional adaptation. These exact structural traits explain why the V4 are so incredibly open to trade, and why they are so sensitive to external shocks. Their inclusion is therefore essential for any comprehensive analysis of global output connectedness.

2 Literature Review

Empirical studies consistently highlight the deep integration of the Visegrád economies into the German industrial core. Since the early 2000s, the V4 region has effectively functioned as the “factory economy” of Europe, relying heavily on German capital and intermediate goods networks [19]. Hagemeyer and Mućk [20] further demonstrate that export-led growth in Central and Eastern Europe is highly sensitive to German domestic demand and export performance. As a result, any economic shock hitting Germany, or passing through it from global markets, hits the V4 economies disproportionately hard.

Beyond this vertical integration with Western Europe, the horizontal ties among the Visegrád Four themselves are equally important. While the V4 countries show highly synchronized business cycles, evidence suggests this co-movement is largely driven by their shared exposure to German supply chains rather than deep bilateral trade with one another, a setup often described as a “hub-and-spoke” network [21] [22]. Still, recent network-based studies have identified real regional connectedness within Central and Eastern Europe. Karkowska and Urjasz [23], for instance, show that shocks in one V4 economy can quickly spill over to its neighbors due to similar institutional setups, geographic proximity, and shared investor sentiment. A key question, therefore, is whether these intra-V4 spillovers are actual bilateral linkages or just a shared reaction to the same external shocks.

Zooming out from Europe, the United States and China naturally exert massive influence over this regional network. The US remains the dominant driver of global business cycles. As Kose et al. [24] show, American growth shocks rapidly propagate to both advanced and emerging markets through trade and financial channels. At the same time, China’s rapid economic rise has fundamentally altered global output. Dizioli et al. [25] find that a slowdown in Chinese growth creates severe negative spillovers globally, particularly punishing export-oriented manufacturing hubs like Germany and, by extension, the V4. Because US consumption, Chinese manufacturing demand, and European industrial output are so tightly linked, measuring these effects requires a framework that can handle multiple countries simultaneously.

To track these complex, shifting linkages, researchers have increasingly moved away from simple spatial autoregressive models or standard static Vector Autoregressions (VAR) [26]. While a full-sample estimation gives a helpful summary of average connectedness, the global economy is constantly changing. Between 1995 and 2023, the world absorbed China’s WTO entry, the 2008 Global Financial Crisis, the European debt crisis,

and the COVID-19 pandemic. Assuming that spillover effects stayed constant across nearly three decades of such turbulence is simply unrealistic.

This is why the dynamic network-based approach introduced by Diebold and Yılmaz [10] [11] has become the methodological standard. By utilizing generalized variance decompositions, this framework isolates pairwise shock contagion without forcing the researcher to assume a specific variable ordering. More importantly, by estimating the VAR model repeatedly over a rolling window of observations, it captures how macroeconomic linkages actually evolve over time [12]. As shown in recent empirical work [27], this yields a time series of indices that track the exact intensity and direction of spillovers. For our study, this capability is essential. It allows us to observe how countries shift between being net transmitters and receivers during specific crises, and to test exactly how strongly US or Chinese shocks ripple through Germany before reaching Central Europe. Ultimately, this rolling-window approach gives a much more realistic and detailed picture of international business cycles than traditional static models.

3 Dataset

The data employed in this study consist of the real Gross Domestic Product (GDP) at constant 2015 U.S. dollars, covering the period from 1995 to 2023. This time frame was selected to ensure broad data availability and to adequately capture both the post-transition economic dynamics in Central and Eastern Europe and the structural developments in major global economies. The countries included in the analysis are the United States, China, Germany, and the Visegrád Four (V4) nations: Czechia, Hungary, Poland, and the Slovak Republic. These economies were selected due to their systemic relevance, deep trade and investment ties, and their increasingly synchronized business cycles over the past three decades.

All core GDP data were retrieved from the World Bank’s World Development Indicators (WDI) database (Indicator: NY.GDP.MKTP.KD). Because quarterly GDP data are not consistently available across the entire sample period for all selected countries, the annual series were converted to a quarterly frequency using linear interpolation. Furthermore, to facilitate comparability across economies of vastly different absolute sizes and to stabilize the variance, the interpolated GDP series were natural log-transformed.

The resulting logarithmic real GDP trajectories are presented in Figure 2. As the figure illustrates, the United States consistently operates on the highest absolute economic scale, maintaining a steady, mature growth trajectory throughout the sample period. In stark contrast, China demonstrates an exceptionally steep convergence path, reflecting rapid economic expansion that bridges the gap toward the US levels. Germany represents the advanced, mid-tier economic core with moderate growth, while the V4 countries move in a highly synchronized bundle at the lower end of the absolute scale, displaying more volatile but steady long-term catch-up dynamics.

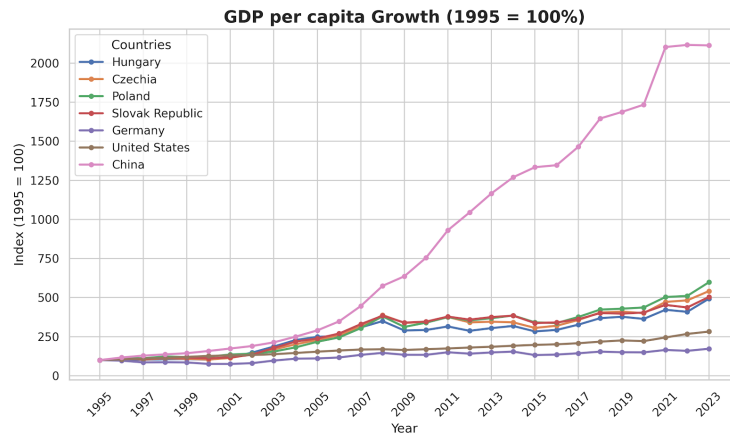


Figure 2: Evolution of Log Real GDP (1995–2023)

In addition to linear interpolation, we also experimented with spline interpolation methods, including quadratic and cubic splines. However, the quarterly series generated by these spline-based methods produced output that was difficult to interpret both in the short and long run, due to unrealistic dynamics and artifacts that did not reflect the underlying economic trends. As a result, the linear interpolation approach was retained for the main analysis.

Standard deviation determines the volatility of a country, and we can see at Table 1 that China shows the highest volatility by far, reflecting its rapid structural transformation and dynamic economic growth while countries like Germany and the United States exhibit lower volatility, indicating more stable and consistent economic development. On the other hand, these volatilities may also reflect how susceptible a country’s economy is to external or internal shocks. Higher volatility could indicate that an economy is more sensitive to global financial conditions, trade fluctuations, or domestic structural imbalances. Conversely, lower volatility may suggest more robust institutions, diversified economic structures, or effective stabilization policies that buffer the impact of such shocks.

The United States and China exhibit the highest mean and median values of log real GDP, reflecting their large absolute economic sizes. Germany also shows high values, while Slovak Republic and Hungary have the lowest means, indicating smaller economies in comparison.

All countries exhibit negative skewness, meaning the distribution of log real GDP is left-skewed, with more observations above the mean and some extreme low values. The Slovak Republic (-0.3111) and the United States (-0.2570) have the most pronounced left skewness, while Poland (-0.0723) is the least skewed, indicating a relatively symmetric distribution. All kurtosis values are below 3, indicating platykurtic distributions, which are flatter than a normal distribution and have fewer and less extreme outliers. Hungary (2.2708) and the United States (2.2416) are closest to normality, while China and the Slovak Republic have the lowest kurtosis (1.66 and 1.63 respectively), suggesting lighter tails. The Jarque-Bera test results show that China, Czechia, Germany, Poland, and the Slovak Republic reject the null hypothesis of normality at the 5% significance level, while Hungary and the United States have p-values above 0.05, indicating no significant deviation from normality.

Overall, most distributions are mildly non-normal with negative skewness and platykurtic characteristics, while Hungary and the United States display log real GDP dynamics that are relatively symmetric and closer to a normal distribution. China’s high standard deviation reflects strong economic transformation and fluctuations consistent with its rapid growth. Countries with lower skewness and kurtosis values, such as Poland and Hungary, may reflect more stable long-term economic trends in logarithmic terms.

Country	Mean	Median	Max	Min	Std. Dev.	Skewness	Kurtosis	Jarque-Bera	JB p-value	Obs
China	29.4661	29.5520	30.4745	28.2361	0.6987	-0.2015	1.6636	9.1732	0.0102	113
Czechia	25.8314	25.8887	26.1175	25.4963	0.2004	-0.2448	1.7234	8.8014	0.0123	113
Germany	28.7837	28.7857	28.9408	28.5915	0.1049	-0.0813	1.8414	6.4441	0.0399	113
Hungary	25.4552	25.4672	25.7834	25.0063	0.1923	-0.2336	2.2708	3.5311	0.1711	113
Poland	26.6842	26.7111	27.1817	26.1119	0.3885	-0.0723	1.8340	6.5000	0.0388	113
Slovak Republic	24.8906	25.0659	25.3715	24.4516	0.2900	-0.3111	1.6371	10.5773	0.0056	113
United States	30.4148	30.4232	30.7249	30.0368	0.1785	-0.2570	2.2416	3.9520	0.1336	113

Table 1: Descriptive statistics for log real GDP

4 Methodology

4.1 GFEVD (Generalized Forecast Error Variance Decomposition)

This thesis aims to investigate and quantify the economic connectedness between the Visegrád Four (Czechia, Hungary, Poland, and the Slovak Republic) and three of the world’s largest economies: the United States, China, and Germany. To achieve this, we use the Generalized Forecast Error Variance Decomposition (GFEVD) framework, which is an essential tool for analyzing how shocks originating in one economy propagate to others through dynamic interactions over time.

While traditional Vector Autoregressive (VAR) models are widely used to assess total spillovers in a system by capturing the general influence of all shocks across variables, they often rely on orthogonalization techniques, such as Cholesky decomposition. This imposes a strong assumption on the ordering of variables, which may be arbitrary and potentially misleading in systems with complex or reciprocal relationships.

In contrast, the GFEVD, developed by Pesaran and Shin [28], allows for a more refined analysis by capturing directional spillovers - that is, the impact of a shock in one specific variable (or country) on another. This approach eliminates the need for variable ordering and provides a more robust basis for measuring interconnectedness in systems characterized by complex economic linkages, as is the case with international trade and capital flows.

Consider a covariance stationary N -variable VAR(p):

$$\mathbf{x}_t = \sum_{i=1}^p \Phi_i \mathbf{x}_{t-i} + \varepsilon_t, \quad \varepsilon_t \sim (0, \Sigma). \quad (1)$$

Here, ε_t is a vector of independently and identically distributed disturbances. The moving average representation of this VAR process is given by:

$$\mathbf{x}_t = \sum_{i=0}^{\infty} \mathbf{A}_i \varepsilon_{t-i}, \quad (2)$$

where the $N \times N$ coefficient matrices $\mathbf{A}_i$ obey the recursion:

$$\mathbf{A}_i = \Phi_1 \mathbf{A}_{i-1} + \Phi_2 \mathbf{A}_{i-2} + \cdots + \Phi_p \mathbf{A}_{i-p}. \quad (3)$$

This moving average representation is fundamental as it allows us to express current values of $\mathbf{x}_t$ in terms of past and present shocks ε_t . Building on this, the GFEVD allows for the decomposition of the H -step-ahead forecast error variance of variable i due to shocks in variable j . This decomposition is expressed as:

$$\theta_{ij}^{(H)} = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' \mathbf{A}_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} (e_i' \mathbf{A}_h \Sigma \mathbf{A}_h' e_i)}, \quad (4)$$

where e_i is a selection vector with 1 in the i -th position and 0 elsewhere, $\mathbf{A}_h$ is the moving average coefficient matrix, σ_{jj} is the standard deviation of the error term for the j -th equation, Σ is the variance matrix for the error vector ε , $\theta_{ij}^{(H)}$ represents the portion of variable i 's forecast error variance at horizon H due to shocks in variable j .

To ensure comparability and interpretation across different variables, the variance shares are normalized such that each row sums to one:

$$\tilde{\theta}_{ij}^{(H)} = \frac{\theta_{ij}^{(H)}}{\sum_{j=1}^N \theta_{ij}^{(H)}}. \quad (5)$$

By construction, $\sum_{j=1}^N \tilde{\theta}_{ij}^{(H)} = 1$. These normalized GFEVD values form the basis of connectedness tables and indices, including total, directional (to/from), and net connectedness.

4.2 Spillover Measures

Based on the normalized generalized forecast error variance decomposition values, various spillover measures can be constructed to quantify the extent and direction of connectedness within the system, as outlined by Diebold and Yilmaz [10].

Total Spillovers

The total connectedness index, or total spillover index, quantifies the contribution of shocks across all variables to the total forecast error variance of the system. It is calculated as the sum of all off-diagonal elements in the normalized GFEVD matrix, representing the portion of the forecast error variance that comes from spillovers:

$$S^g(H) = \frac{\sum_{i,j=1, j \neq i}^N \tilde{\theta}_{ij}^g(H)}{N}. \quad (6)$$

This index measures the aggregate interdependency within the system.

Directional Spillovers

While the total volatility spillover index suffices for understanding the overall magnitude of shocks transmitted across major asset classes, the generalized VAR approach is essential for discerning the direction of these volatility spillovers. Since the generalized impulse responses and variance decompositions are robust to variable ordering, the normalized elements of the generalized variance decomposition matrix provide a reliable basis for calculating these directional spillovers. Directional spillovers capture the impact of shocks from one variable to another, providing a detailed view of the network's dynamics.

- **Directional Spillovers *From*** (spillovers received): This measures the total spillovers variable i receives from all other variables j :

$$S_{i.}^g(H) = \frac{\sum_{j=1, j \neq i}^N \tilde{\theta}_{ij}^g(H)}{N}. \quad (7)$$

- **Directional Spillovers *To*** (spillovers transmitted): This measures the total spillovers variable i transmits to all other variables j :

$$S_{.i}^g(H) = \frac{\sum_{j=1, j \neq i}^N \tilde{\theta}_{ji}^g(H)}{N}. \quad (8)$$

Net Spillovers

The net spillover from variable i is the difference between the spillovers it transmits to other variables and the spillovers it receives from other variables. A positive value indicates that variable i is a net transmitter of shocks, while a negative value indicates it is a net receiver.

$$S_i^g(H) = S_{.i}^g(H) - S_{i.}^g(H). \quad (9)$$

Thus, while simple VAR models are useful for summarizing overall system-wide spillovers, the generalized framework allows us to uncover more granular, directional interdependencies, a crucial feature when analyzing how the Visegrád economies interact with, influence, or are influenced by, major global players in the context of external economic shocks.

5 Preliminary Statistical Testing and Model Selection

The statistical analyses for the Generalized Forecast Error Variance Decomposition (GFEVD) model were developed with inspiration from Obuabor’s (2023) research on output connectedness [29]. However, this study applies the methodology to different economic blocs, reflecting the distinct scope and focus of the current analysis.

5.1 Unit Root Tests (ADF and Phillips-Perron)

To ensure that the underlying assumptions for vector autoregressive (VAR) modeling are satisfied, we first tested the stationarity of each country’s log real GDP time series using the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests. The presence of unit roots in the level data would indicate non-stationarity, potentially biasing model estimates unless appropriate transformations are applied.

In levels, both tests suggest that all series are non-stationary. For instance, the ADF test yields p-values well above 0.05 for Germany (0.6725), China (0.4508), and others. Similar conclusions emerge from the PP test, confirming that the level series follow unit root processes.

Country	ADF Test	PP Test
China	-1.6620 (p = 0.4508)	-2.5518 (p = 0.1034)
Czech Republic	-1.0871 (p = 0.7202)	-0.8963 (p = 0.7892)
Germany	-1.2025 (p = 0.6725)	-1.2473 (p = 0.6529)
Hungary	-0.9742 (p = 0.7625)	-0.8646 (p = 0.7994)
Poland	-0.4148 (p = 0.9076)	-1.6561 (p = 0.4539)
Slovakia	-1.0634 (p = 0.7295)	-1.6577 (p = 0.4531)
USA	-0.8400 (p = 0.8071)	-1.6948 (p = 0.4338)

Table 2: ADF and Phillips-Perron Tests (Non-Differenced)

After differencing once, however, most of the series become stationary. The first-differenced ADF and PP statistics for countries such as Czechia (ADF p = 0.0400) and Germany (PP p = 0.0135) clearly reject the null hypothesis of a unit root. This validates the use of the first-differenced series in the subsequent VAR framework, ensuring statistical consistency and avoiding spurious regression problems.

Country	ADF Test	PP Test
China	-1.0117 (p = 0.7490)	-2.1057 (p = 0.2422)
Czech Republic	-2.9480 (p = 0.0400)	-3.3869 (p = 0.0114)
Germany	-2.7038 (p = 0.0734)	-3.3334 (p = 0.0135)
Hungary	-1.8453 (p = 0.3582)	-3.3872 (p = 0.0114)
Poland	-2.6349 (p = 0.0860)	-2.9164 (p = 0.0435)
Slovakia	-2.3830 (p = 0.1466)	-3.3382 (p = 0.0133)
USA	-2.8403 (p = 0.0527)	-3.4509 (p = 0.0093)

Table 3: ADF and Phillips-Perron Tests (Differenced)

5.2 Lag length selection

Selecting an appropriate lag length is crucial for capturing the dynamic interactions among the variables without overfitting. We relied on four commonly used selection criteria: Akaike Information Criterion (AIC), Final Prediction Error (FPE), Hannan-Quinn Information Criterion (HQIC), and Schwarz Bayesian Criterion (BIC).

All four criteria point to an optimal lag length of 12 (Table 6). Given the quarterly frequency of the data, this implies that up to three years of lags are needed to adequately explain the intertemporal relationships in the system. This relatively long lag structure is justified by the complexity and potential delays in macroeconomic spillovers between large and diverse economies such as the United States, China, Germany and the Visegrád countries.

Although these information criteria can be derived through specific equations, in this study I relied on a Python implementation to determine the optimal number of lags, using a specification with a constant term.

5.3 Johansen Cointegration Test

To evaluate whether the non-stationary level series exhibit long-run equilibrium relationships, we conducted the Johansen cointegration test. Surprisingly, both the trace and maximum eigenvalue statistics suggest that the rank of the cointegration matrix Π is 7, which equals the total number of variables in the system (Table 7).

To evaluate whether the non-stationary level series exhibit long-run equilibrium relationships, we conducted the Johansen cointegration test. Surprisingly, both the trace and maximum eigenvalue statistics suggest that the rank of the cointegration matrix Π is 7, which equals the total number of variables in the system.

A full-rank cointegration matrix implies that all variables are stationary in levels. This result contradicts our earlier unit root tests, which indicated that the individual series are integrated of order one ($I(1)$). Such a discrepancy may arise due to small-sample distortions, lag selection issues, or model misspecification.

5.4 Stability Diagnostics

We tested the dynamic stability of the VAR model by examining the inverse roots of the companion matrix. For a VAR to be dynamically stable, all roots must lie strictly inside the unit circle. In our model, some roots slightly exceed this boundary (e.g., 1.0051, 1.0150 at Table 8), implying that the system is not strictly stable in the mathematical sense, which can be seen at Figure 3.

This mild instability can likely be attributed to the relatively large number of lags (12), which increases the complexity of the model and can result in marginal violations of the stability condition. Nevertheless, such small deviations are common in high-dimensional VARs with long lags and do not necessarily undermine the

interpretability of generalized forecast error variance decompositions (GFEVD), which are robust to moderate instability. However, this finding calls for cautious interpretation of the dynamic results.

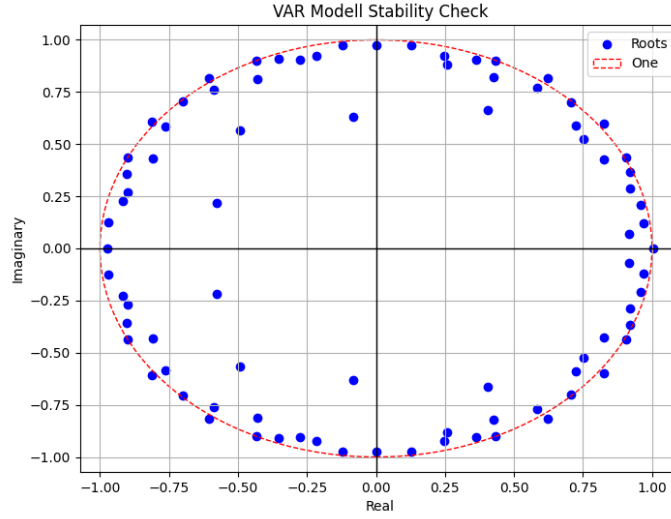


Figure 3: Stability

5.5 Residual Autocorrelation: Ljung-Box Test

To assess whether the chosen lag structure adequately captures the temporal dependencies in the data, we applied the Ljung-Box test to the residuals of each country's equation. The results uniformly indicate no significant residual autocorrelation at lag 12. For example, the p-values for Czechia (0.813) and the United States (0.938) are well above conventional significance thresholds.

These findings suggest that the model effectively captures the dynamics of each time series and that no systematic information is left in the residuals. The absence of autocorrelation supports the adequacy of the selected lag length and enhances the reliability of our forecast-based connectedness analysis.

Country	LB Statistic	p-value
China	4.81	0.964
Czech Republic	7.63	0.813
Germany	8.17	0.772
Hungary	13.95	0.304
Poland	15.64	0.208
Slovakia	9.34	0.674
USA	5.52	0.938

Table 4: Ljung-Box Test Results (lag = 12)

5.6 Normality of Residuals

Finally, we examined the residuals for deviations from normality using the Jarque-Bera and Shapiro-Wilk tests. Strong evidence of non-normality was found in several countries, particularly China and Germany. For instance, China’s Jarque-Bera statistic is 575.01 with a p-value near zero, indicating significant skewness and excess kurtosis.

Similarly, the Shapiro-Wilk test reveals substantial departures from normality ($W = 0.7860$ for China, $p < 9.42 \cdot 10^{-11}$). While the assumption of normally distributed residuals is important for certain types of inference, generalized forecast error variance decompositions do not rely on this condition. Thus, although the results suggest non-Gaussian shocks, the core methodology of the connectedness analysis remains valid.

Country	JB Stat.	JB p	SW Stat.	SW p
China	575.01	$1.37 \cdot 10^{-125}$	0.7860	$9.42 \cdot 10^{-11}$
Czech Republic	88.96	$4.82 \cdot 10^{-20}$	0.8866	$3.48 \cdot 10^{-7}$
Germany	538.60	$1.11 \cdot 10^{-117}$	0.7695	$3.18 \cdot 10^{-11}$
Hungary	19.56	$5.66 \cdot 10^{-5}$	0.9514	0.0010
Poland	312.94	$1.11 \cdot 10^{-68}$	0.8334	$3.03 \cdot 10^{-9}$
Slovakia	33.19	$6.21 \cdot 10^{-8}$	0.9355	1.04×10^{-4}
USA	100.51	$1.50 \cdot 10^{-22}$	0.9001	$1.43 \cdot 10^{-6}$

Table 5: Normality Tests (Jarque-Bera and Shapiro-Wilk)

6 Results

In the analysis, a forecast horizon of $H = 20$ quarters was chosen for the GFEVD matrix. This horizon corresponds to approximately five years, representing a medium-term perspective on output connectedness. Selecting this horizon balances capturing both short-term dynamics and longer-term spillover effects, providing a comprehensive view of how shocks propagate across economies over a meaningful timeframe without extending into overly uncertain long-term predictions.

The sample size was not further disaggregated due to methodological constraints. According to general econometric considerations in the VAR framework, a reliable Generalized Forecast Error Variance Decomposition (GFEVD) typically requires a minimum sample of approximately 100 [11] observations, especially when modeling systems with several variables and longer forecast horizons. Further increasing the frequency (e.g., by using more aggressive interpolation or higher-frequency proxies) would reduce the effective degrees of freedom and potentially undermine the statistical reliability and interpretability of the connectedness estimates.

The analysis of connectedness and spillover indicators across the examined countries reveals a clear structure of global and regional interdependence within the economic (or financial) network, which is shown at Figure 4.

The spillover heatmap offers further insights into bilateral transmission patterns. It reveals that China has high spillover values across several directions, especially toward Slovakia (0.21) and the United States (0.18), reflecting China’s global economic footprint. Interestingly, the most intensive bilateral spillovers are not necessarily regional but global - for example, the linkages between China–USA, China–Slovakia, and Germany–Poland are notably strong. This underscores that shock transmission is shaped not only by geographic proximity but by deeper patterns of global economic interdependence.

The United States shows a more balanced profile. With the received and the transmitted values nearly equal (0.89 vs. 0.83), its net connectedness is only slightly positive (0.06). This suggests that the U.S. plays a dual role as both a major transmitter and receiver of shocks, characteristic of a globally integrated economy.

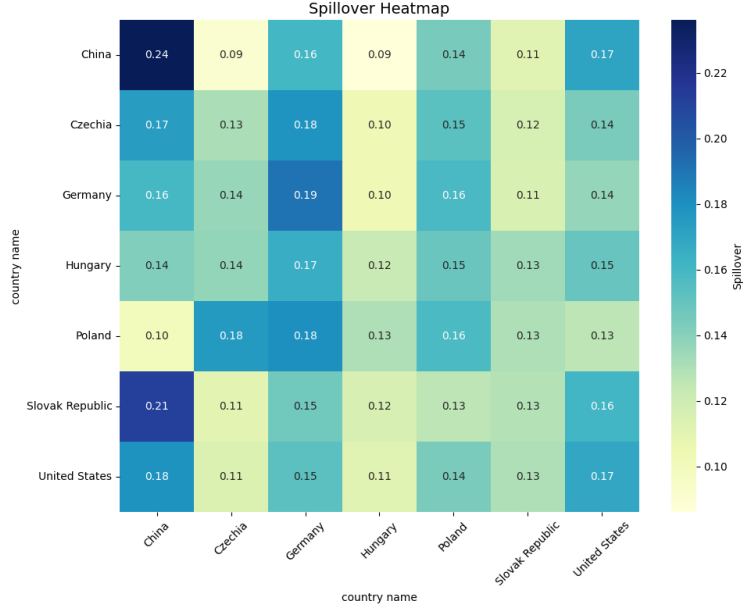


Figure 4: Volatility Spillover

According to Figure 5, China and Germany emerge as the most influential net shock transmitters within the system. China's high "to connectedness" value (0.97) and positive net effect (0.20) indicate that it exerts a substantial influence on other countries while being relatively less exposed to external shocks. Germany displays a similar pattern with an even higher transmitted value (0.99) and a net connectedness of 0.17, suggesting that both countries play a leading role in propagating economic shocks globally and are structurally central to the network.

In contrast, the Visegrád countries - particularly Hungary, Slovakia, and Czechia - act more as net recipients of shocks. Hungary shows the most negative net connectedness (-0.23), while Slovakia (-0.15) and Czechia (-0.09) also have clearly negative values. This highlights that these smaller, open economies are more vulnerable to external influences than they are able to influence others, reflecting their economic integration and dependence on larger global players like Germany and China.

In summary, the findings highlight a global structure in which China and Germany are key transmitters of economic shocks, the United States holds a balanced position, and the Visegrád Four countries are structurally positioned as net recipients, emphasizing their sensitivity to external developments and their embeddedness within a broader international system.

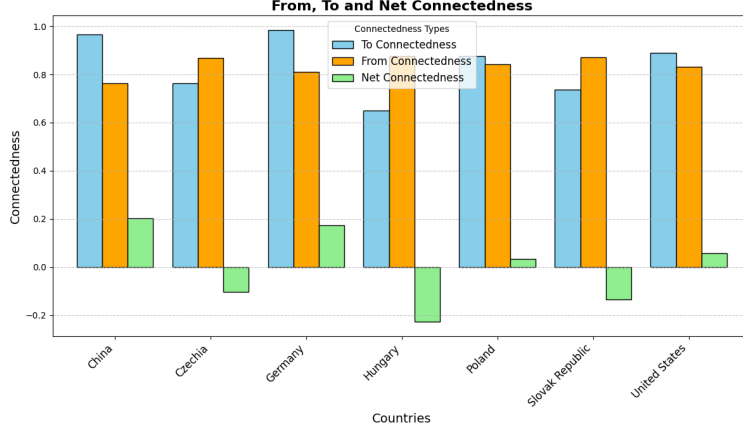


Figure 5: Net Connectedness

A closer look at the Visegrád Four countries - Czechia, Hungary, Poland, and Slovakia - reveals important nuances in their systemic roles.

While three out of the four countries exhibit negative net connectedness, the degree varies. Hungary stands out with the lowest net value (-0.23), signaling its heightened vulnerability to external shocks, especially those originating from Germany and China. Slovakia (-0.15) and Czechia (-0.09) also show clear patterns of being net receivers, while Poland has a marginally positive net connectedness (0.03), suggesting a slightly more balanced role in the network.

From the spillover matrix, Poland and Czechia exhibit relatively strong bilateral connections with Germany and among each other, consistent with their tight integration into regional supply chains, especially in the automotive and manufacturing sectors. Slovakia's high spillover reception from China (0.21) may reflect sectoral dependencies or external trade structures.

Overall, the V4 countries do not function as major drivers of economic or financial spillovers, but rather as responsive nodes within a network dominated by larger economies. Their connectedness structure reflects their economic openness, regional integration (especially with Germany), and exposure to global market trends. This structural positioning makes coordinated regional policy responses and resilience-building efforts particularly important for the V4 bloc.

7 Discussion

The empirical findings of this study provide profound insights into the structural dynamics of global output connectedness. By mapping the real GDP spillovers among the United States, China, Germany, and the Visegrád Four, the results corroborate the hypothesis that the modern macroeconomic landscape is characterized by severe structural asymmetries. Rather than operating as isolated entities, these economies form a hierarchical transmission network where shocks cascade from global originators through regional hubs to highly exposed peripheral receivers.

First, the results firmly validate the theoretical conceptualization of the Visegrád Four as Dependent Market Economies (DMEs), as proposed by Nölke and Vliegenthart [14]. The consistently negative net connectedness of Hungary (-0.23), Slovakia (-0.15), and Czechia (-0.09) empirically quantifies their structural vulnerability. As "factory economies" deeply embedded in European supply chains, their growth models are highly elastic to external demand, making them systemic shock absorbers rather than originators. However, the marginal positive net connectedness of Poland (0.03) presents a notable nuance. Unlike its regional peers, Poland benefits from a significantly larger domestic market, which acts as a macroeconomic buffer against external demand fluctuations, affording it a slightly more resilient and balanced position within the network.

Second, the pivotal role of Germany as a central transmission hub is unequivocally supported by the data. With an outstandingly high "to connectedness" value (0.99) and a robust positive net effect (0.17), Germany bridges the gap between global economic shifts and Central European industrial output. The strong bilateral spillovers observed between Germany and the V4 nations confirm the "hub-and-spoke" architecture described in the literature [21]. Shocks originating in the US or China do not simply bypass Western Europe; they are absorbed, amplified, and subsequently transmitted to the V4 through the German manufacturing and export apparatus.

Third, the results capture a broader shift in global economic influence. China emerges as the most dominant net transmitter of output shocks in the system (net connectedness of 0.20 and a "to" index of 0.97). This reflects China's rapid structural transformation over the 1995-2023 period, cementing its status not merely as a global assembly line, but as a primary originator of systemic demand and supply shocks. The surprisingly strong direct spillovers from China to Slovakia (0.21) highlight that modern global value chains can circumvent traditional geographic proximity. Meanwhile, the United States displays a highly balanced profile (net connectedness of 0.06). As the world's largest consumer and financial market, the US absorbs nearly as much real output variance from the global system as it transmits, reflecting the mature, dual nature of its global integration.

From a methodological standpoint, the study highlights both the strengths and the challenges of modeling complex, multi-country macroeconomic networks. While the required first-differencing of the data and the mild dynamic instability inherent in the 12-lag VAR structure pose certain inferential challenges, the Generalized Forecast Error Variance Decomposition (GFEVD) framework proved robust. The absence of residual autocorrelation confirms that the chosen lag structure successfully captured how these output shocks cross borders over time, even when the underlying data does not follow a perfectly normal distribution.

From a practical standpoint, this extreme reliance on German demand leaves the V4 highly vulnerable during global downturns. To build economic resilience, Central European governments need to look beyond intermediate manufacturing, diversify their export destinations, and encourage stronger domestic consumption to help absorb the impact of global market swings.

8 Conclusion

The primary objective of this study was to trace how real economic output shocks traveled among the United States, China, Germany, and the Visegrád Four from 1995 to 2023. By applying the generalized variance decomposition framework developed by Diebold and Yilmaz, we were able to measure both the scale and the exact direction of these macroeconomic spillovers. Rather than just confirming that these economies are connected, the data provides a clear picture of who drives the global business cycle and who merely reacts to it.

A central takeaway is the distinctly vulnerable position of the Visegrád Four. With the slight exception of Poland, the V4 nations operate almost exclusively as shock absorbers in the global network. Because their growth models rely so heavily on foreign capital, functioning largely as Dependent Market Economies deeply tied to German supply chains, they are highly exposed to outside economic shifts. They rarely generate systemic shocks themselves; instead, they import the volatility created by larger global players.

This dynamic highlights a difficult reality for Central Europe: the region's economic performance cannot be separated from the health of the major global powers. The extreme openness of the V4 has undoubtedly fueled their historical catch-up growth, but today, amid shifting trade routes and recurring global crises, it also acts as a profound vulnerability. Moving forward, it would be valuable to break these spillovers down by specific industries, such as automotive or high-tech manufacturing, and to use time-varying models to track exactly how these connections behave during specific market downturns. From a policy perspective, the conclusion is straightforward. While deep economic integration remains vital, sustained growth in the V4 will increasingly depend on building stronger domestic macroeconomic resilience to cushion against the inevitable swings of the global market.

Appendix

Lag	AIC	BIC	FPE	HQIC
0	-76.96	-76.78	3.77×10^{-34}	-76.89
1	-85.07	-83.61	1.14×10^{-37}	-84.48
2	-84.33	-81.60	2.40×10^{-37}	-83.23
3	-83.77	-79.76	4.36×10^{-37}	-82.15
4	-83.78	-78.49	4.63×10^{-37}	-81.64
5	-85.48	-78.92	9.50×10^{-38}	-82.83
6	-84.85	-77.01	2.15×10^{-37}	-81.68
7	-84.57	-75.45	3.72×10^{-37}	-80.88
8	-87.12	-76.73	4.31×10^{-38}	-82.92
9	-90.61	-78.94	2.33×10^{-39}	-85.89
10	-90.31	-77.37	7.16×10^{-39}	-85.07
11	-90.84	-76.61	1.46×10^{-38}	-85.08
12	-111.7	-96.20	9.07×10^{-47}	-105.4

Table 6: VAR Lag Order Selection

Rank	Trace Statistic	95% Critical Value	99% Critical Value
0	5431.98	111.78	121.74
1	3466.80	83.94	92.71
2	1936.91	60.06	67.64
3	815.41	40.17	46.57
4	236.33	24.28	29.51
5	145.55	12.32	16.36
6	55.69	4.13	6.94

Table 7: Johansen Cointegration Test - Trace Statistic

Index	Root	Index	Root
1	0.6183	22	0.9715
2	0.6183	23	0.9728
3	0.6369	24	0.9735
4	0.6369	25	0.9744
5	0.7504	26	0.9747
6	0.7504	27	0.9782
7	0.7757	28	0.9783
8	0.7757	29	0.9796
9	0.9158	30	0.9801
10	0.9158	31	0.9830
11	0.9160	32	0.9903
12	0.9160	33	0.9932
13	0.9186	34	0.9964
14	0.9186	35	0.9992
15	0.9189	36	0.9998
16	0.9189	37	1.0002
17	0.9191	38	1.0037
18	0.9191	39	1.0051
19	0.9245	40	1.0150
20	0.9245	41	1.0151
21	0.9316	42	1.0205

Table 8: Inverse Roots of the Characteristic Polynomial

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